

Milestone 1 Findings

Literature review · framework assessment · data compendium

A clean source-led deck: every finding uses an excerpt rendered directly from the cited paper, report, or official page.

Seven themes

What the primary literature converges on



LITERATURE REVIEW

SOURCE-LED FINDING

No existing framework was designed for TRL 1–3

ORIGINAL SOURCE EXCERPT · p. 26

ambitious vision for radically new technology, with potential to create new markets, respond to societal needs and/or to provide solutions for global challenges. EIC Pathfinder Open supports early-stage development of such future technologies (e.g., various activities at low Technology Readiness Levels from 1 to 4), based on high-risk/high-gain science-towards-technology breakthrough research (‘deep tech’). This research must provide the foundations of the technology you are envisioning.

EIC Pathfinder Open may support your work, especially if it is highly risky: you may set out to try things that will not work; you may be faced with questions that nobody knows the answer to yet; you may realise that there are many aspects of the problem that you

26

European Commission (2026), EIC Work Programme 2026, p. 26 (EIC Pathfinder).

COMMENTS

- The space where NCCR Muoniverse operates — fundamental physics at TRL 1–3 — is precisely where no commercialization framework was designed to work.
- Every framework assessed (RCA/Imperial, CRL, Innosuisse, EIC) has a structural gap at this stage.
- This gap is the thesis's core motivation: identify what support mechanisms fill it.

LITERATURE REVIEW

SOURCE-LED FINDING

Ecosystem, not policy, drives deep tech

ORIGINAL SOURCE EXCERPT · p. 1

SPRINGER NATURE Link

Building deep-tech entrepreneurs: factors that promote the presence of startups in European regions

Research Open access Published: 09 April 2026

COMMENTS

- The primary study finds that specific deep-tech knowledge and talent matter more for startup emergence than generic ecosystem quality or formal institutions.
- Its regional analysis covers 272 European NUTS-2 regions and 1,605 deep-tech startups.
- → For Switzerland, specialized research infrastructure and talent concentration are strategic assets.

Micol, F., Leendertse, J. & van Rijnsoever, F. (2026), Journal of Technology Transfer 51, pp. 2886–2915.

LITERATURE REVIEW

SOURCE-LED FINDING

TTOs: relationships beat transactions

ORIGINAL SOURCE EXCERPT · p. 16

from R&D (Mazzucato & Semmler, 2017; Pisano, 2000). Therefore, debate about how to fund innovation is important – particularly innovation that addresses grand challenges affecting environment (5H1) and civil society (5H2) such as clean energy, quality of life, food security, and climate change, which require transformative (i.e., disruptive and risky) approaches unlikely to be funded from private sources.

6 Conclusion

Addressing the calls to investigate technology transfer within the university innovation service ecosystem more holistically (de Falani Bezerra & Torkomian, 2023; Good et al., 2019; Miller et al., 2018), this study emphasises the importance of a wider view of quintuple helix (5H) stakeholders including society and the environment. This study extends current knowledge about how TTOs facilitate stakeholder relationships through the technology commercialization process, by integrating university competencies and market demand. The research highlighted that TTOs’ power is increasing, as research follows funding. TTOs’

COMMENTS

- Chen et al. describe TTOs as actors that translate science into commercially and socially valuable outcomes.
- The playmaker role is relational: TTOs build relationships across the wider university innovation-service ecosystem.
- → PSI's TTO should be designed as an active orchestrator, not only a transaction desk.

Chen, Z., Little, V. & Thuan, N. (2025), Journal of Technology Transfer 50, pp. 1060–1079.

LITERATURE REVIEW

SOURCE-LED FINDING

Structured frameworks work — if adapted

ORIGINAL SOURCE EXCERPT · p. 1

and networks required to navigate commercialization pathways effectively. In response, a new generation of government-supported programs has emerged globally, with a primary focus on early-stage entrepreneurial development. This study examines the origins, design, and evolution of such initiatives across eight countries, most of which are modeled on the U.S. National Science Foundation’s Innovation Corps (I-Corps) program but have been adapted to reflect regional contexts and institutional needs. Through cross-case analysis, we demonstrate that these programs constitute a multi-tiered infrastructure that strengthens the translation of academic research into market-ready innovations. These initiatives are increasingly vital to the advancement of entrepreneurial universities and the maturation of national and regional innovation ecosystems. The paper concludes with comparative insights and suggestions for future research. For policymakers and program designers, the findings highlight how program choices can be used as levers to strengthen entrepreneurial capacity, align interventions with the diverse needs of scientists, and improve the integration of customer discovery and related forms of training withing broader ecosystems.

Keywords Academic entrepreneurship · Commercialization · University spinoffs · Lean

COMMENTS

- I-Corps (lean + customer discovery, 7-week immersion) independently adopted in 8 countries — a natural experiment: structured works, but every system adapted it locally.
- Frameworks need explicit abandon/pivot decision points — sunk-cost protection on 5–15 yr timelines.
- 100 customer interviews in 9 weeks = paradigm shift in scientists' market understanding.

Hayter, C. S. et al. (2025), Journal of Technology Transfer, DOI 10.1007/s10961-025-10288-6.

LITERATURE REVIEW

SOURCE-LED FINDING

Configuring support matters more than providing it

ORIGINAL SOURCE EXCERPT · p. 32

QB	Prob≥Q	I ² (%)	τ ²
5.08	0.0000	85.20	0.0263
5.56	0.059	89.35	0.0434
5.62	0.0204	58.96	0.0108
5.84	0.0033	54.38	0.0074
6.12	0.878		
6.35	0.0000	78.74	0.0272
6.19	0.0000	90.18	0.0266
7.62	0.0000	85.72	0.0268
8.16	0.173		
8.23	0.0000	91.67	0.0397
8.05	0.0211	43.63	0.0051
8.14	0.247		
6.88	0.0000	90.07	0.0363
6.51	0.0058	53.01	0.0067
6.14	0.200		
5.92	0.0000	86.20	0.0279
5.19	0.4843	0.00	0.0000
6.14	0.247		
6.88	0.0000	90.07	0.0363
6.51	0.0058	53.01	0.0067

COMMENTS

- 3 effective configurations of 5 orchestration functions — no single configuration is universally optimal.
- Founders value mentors (64%) and customer intros (61%) above investment (37%) — capital is NOT Europe's scarcest resource.
- Networking helps ($r = 0.071$); more mentoring can even reduce performance (opportunity cost).

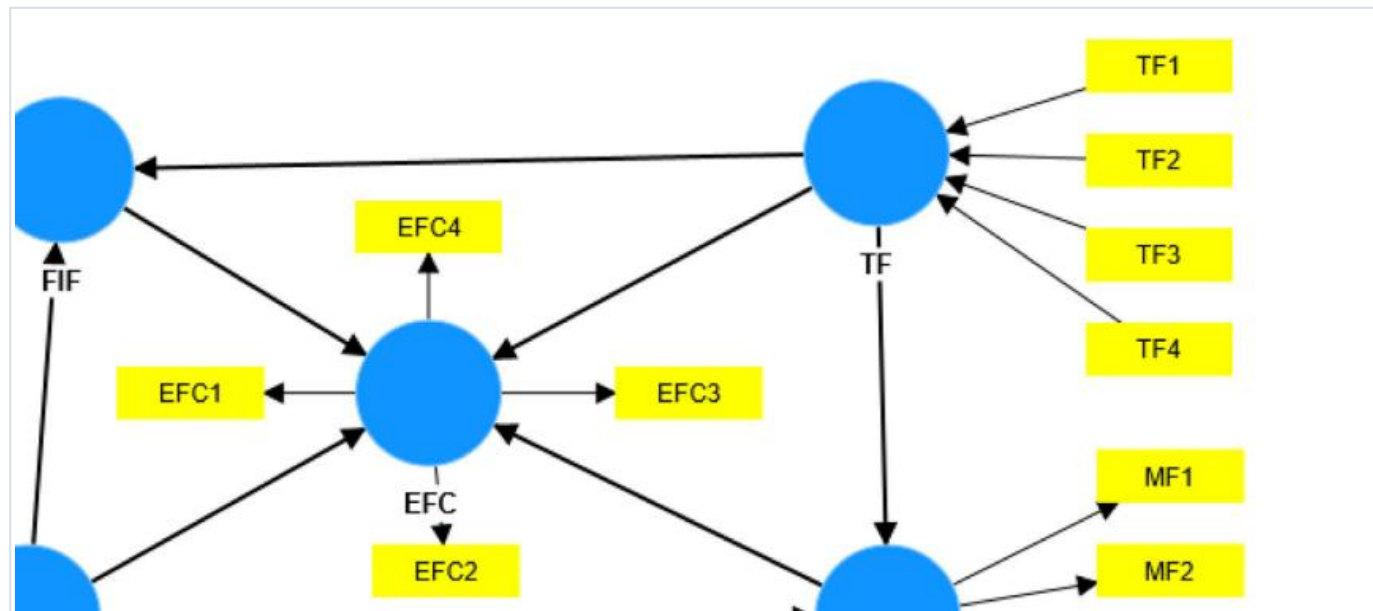
Seitz, N. et al. (2026), Review of Managerial Science, DOI 10.1007/s11846-026-01017-w, Table of moderator effects.

LITERATURE REVIEW

SOURCE-LED FINDING

Market is the binding constraint

ORIGINAL SOURCE EXCERPT · p. 10



Capatina, A., Bleoju, G. & Kalisz, D. (2024), Journal of Innovation & Knowledge 9, 100515.

COMMENTS

- Market features ($\beta = 0.514$) predict success nearly 2× better than technology features ($\beta = 0.310$).
- Firm internal factors (team quality, management) are NOT significant.
- Cantner (n = 1,149): the commercial sphere dominates from day one of spin-off creation.
- → For muons: finding applications beats scientific excellence.

LITERATURE REVIEW

SOURCE-LED FINDING

Commercial exposure from day one

ORIGINAL SOURCE EXCERPT · p. 2

tributing to the formation and growth of new industries, and addressing grand societal challenges (Fini et al., 2018, Rasmussen et al., 2020, Vincett, 2010). However, despite the increasing number of ASOs in recent decades (Mathisen & Rasmussen, 2019), the rate of ASO projects that have failed or been abandoned at some point in the venture creation process remains high, leaving a large stock of knowledge and commercial opportunities unexploited (e.g., Braunerhjelm, 2007; Fini et al., 2017).

Extensive research has been conducted to understand the ASO creation process, focusing on its phases and the barriers encountered along the way. It has been

and commercial spheres represent activities, and social behavior from which derive degree (Merton, 1936) pass the different scientists and their co describe the two embedded during dedness refers to t



COMMENTS

- Dominance analysis: commercial sphere (industry contacts, market knowledge) dominates from the very outset — not in later stages.
- Challenges the assumption 'research first, commercialize later' — empirically wrong.
- → Support design principle: start commercial exposure immediately.

Cantner, U. et al. (2024), Small Business Economics 62, pp. 1555–1590.

LITERATURE REVIEW

SOURCE-LED FINDING

Advisor type matters more than quantity

ORIGINAL SOURCE EXCERPT · p. 19

In our main analyses, we used entropy balancing to correct for intrinsic differences between serial and novice entrepreneurs. As an alternative scheme, we used incorporation at the time of the first survey as the treatment, reasoning that founders who legally incorporate may be more serious and would expend more effort in growing them. We instrument the selection with the same covariates used in the primary analysis. This treatment (Appendix Tables 4, 5) leads to two differences in our findings. First, while still negative, the magnitude of the effect is smaller.

³ Like the coefficients of advice, those of experience reflect standardized units. The magnitude of effect discussed here is for their unstandardized values.



COMMENTS

- Each business advisor: +0.23 application-readiness units (p = 0.04).
- Each technology advisor: -0.31 units (p = 0.05) — actively delays.
- 4 business advisors ≈ +1 full unit ≈ 15 years of entrepreneurial experience.
- → Counterintuitive but evidence-based: limit tech advisors, prioritize business advisors.

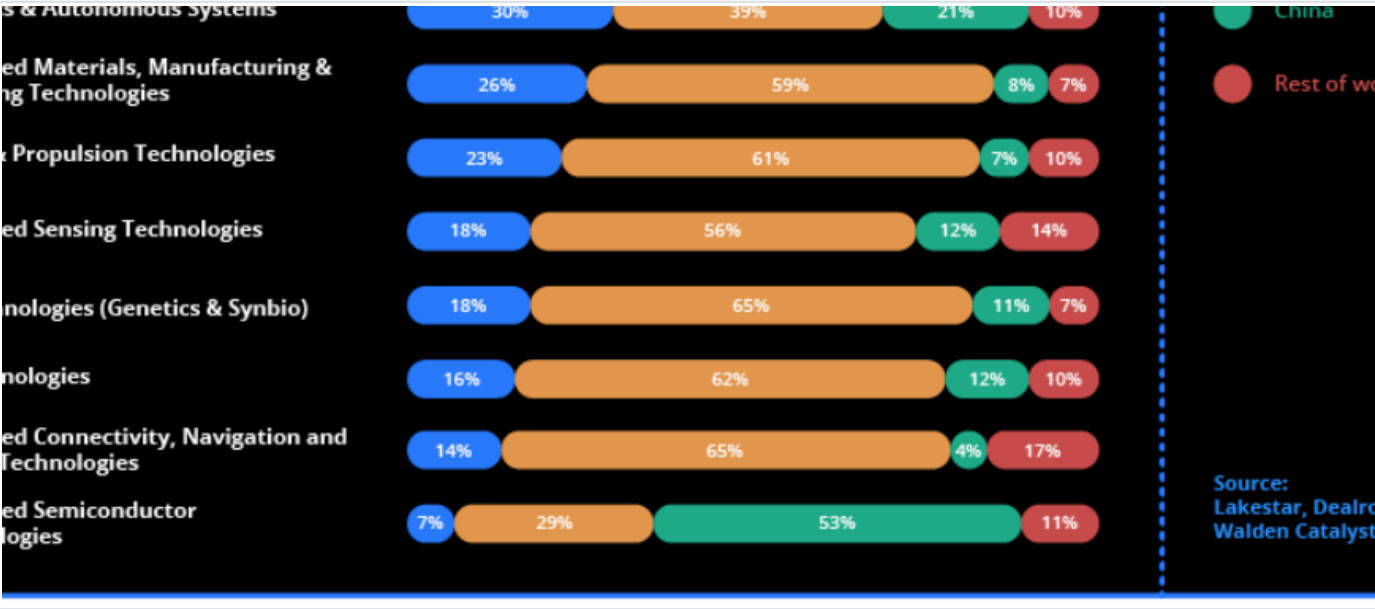
Denoo, L., Van Boxstael, A. & Belz, A. (2024), Journal of Technology Transfer 49, pp. 1567–1605.

LITERATURE REVIEW

SOURCE-LED FINDING

Europe's structural barriers

ORIGINAL SOURCE EXCERPT · p. 3



COMMENTS

- €58B EU deep tech VC vs €215B US — a 3.7× gap.
- 62% of EU investors reject failed founders vs 23% US; 15% vs 45% of PhDs pursue startups.
- 12% vs 40% of corporations engage with deep tech.
- → Cultural and structural — cannot be fixed by capital alone.

Stryber (2025), Deep Tech commercialization in Europe: the gap, the challenge, and the fix.

LITERATURE REVIEW

SOURCE-LED FINDING

Deep tech needs 35% more time, 48% more capital

ORIGINAL SOURCE EXCERPT · p. 54

is the structural gap in funding mechanisms. While VC remains the dominant
ny deep tech startups struggle to secure early-stage financing, particularly for
jects like space tech and advanced computing. Non-dilutive funding sources,
t grants, can play a key role in supporting early innovation, but accessing these
plex and highly competitive, as some stakeholders highlighted in the interviews.
tion process can involve administrative burdens, rigid milestone-based

COMMENTS

- Deep tech takes 35% more time and 48% more capital to reach equivalent milestones.
- These compound with European structural deficits: 3–5 yr VC fund cycles clash with deep tech timelines.
- → Muon tech's 10–20 yr horizon exceeds every Swiss instrument's design parameters.

UNDP (2025), Global Deep Tech Ecosystems: Catalyzing Innovation for Sustainable Development, p. 54.

LITERATURE REVIEW

SOURCE-LED FINDING

Converging on what deep tech is

ORIGINAL SOURCE EXCERPT · p. 1

(staged financing strategies, dual scientific and commercial n multidisciplinary broadening of teams) and the ecosystem level (mu specialized incubation support, and industrial de-risking and scaling parti framework to delineate the boundaries of deep-tech innovation, disting concepts, and propose an agenda for future research.

Keywords: Deep-tech, deep-tech innovation, deep-tech venture, deep-tec

COMMENTS

- 12 defining attributes across 3 levels: technology (science-based, hardware, platforms), venture (5–15+ yr timelines, capital intensity), ecosystem (knowledge, talent, infrastructure).
- Cabanes: deep tech as a 'performative innovation regime' — patient capital, long-term partnerships, ecosystem orchestration.
- → Definitional convergence legitimizes a physics-specific commercialization lens.

Kortsch, J. et al. (2024), Deep-Tech Innovation: A Multi-Method Study, MIT Sloan / University of Münster.

Four frameworks, one gap

RCA / Imperial · CRL · Innosuisse · EIC · candidates



FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

The official pipeline starts at TRL 1 — but scale-up begins later

ORIGINAL SOURCE EXCERPT · p. 1

Develop and scaleup innovations

Develop and scaleup innovations with the potential to create or disrupt markets (TRL 6-8). Patient capital principle.

Startups and SMEs

Single start-ups and individuals intending launch a SME and si mid-caps (can apply equity only).

COMMENTS

- RCA/Imperial: designed for TRL 4–6. CRL: conceptually 1–9 but operationally thin. Innosuisse: optimized for TRL 4–7. EIC Accelerator: TRL 5+.
- The space where Muoniverse operates is where no framework was designed to work.
- → Thesis contribution: adapt/combine existing frameworks to reach this unaddressed space.

European Innovation Council (2026), EIC Accelerator official programme page.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

RCA/Imperial: a cluster, not a program

ORIGINAL SOURCE EXCERPT · p. 1

DT Prime is particularly suited for staff-led ventures that require time, technical validation, and strategic support to reach commercial viability. Built around the specific demands of life science and deep tech, DT Prime management works with Imperial's researchers to provide a combination of non-dilutive funding and tailored support from experienced venture builders.

Every engagement is bespoke, with non-dilutive funding of £50,000–£150,000 in each phase deployed toward clearly defined value inflection points - from generating technical data required by investors and shaping an IP position, to validating market demand or preparing for investor conversations. High-performing teams return for multiple rounds, with cumulative support of up to £500,000.

Science and technology focus areas

We support ventures tackling complex, high-value problems where deep science can unlock new markets, reshape industries, and deliver lasting commercial and societal value.

Imperial College London, DT Prime, Institute for Deep Tech Entrepreneurship (official page).

COMMENTS

- 4 components: Innovation Challenge (1-week MBA), Wicked Module (1-month, with Steve Blank 2023), Wicked Acceleration Labs (WAL), DT Prime (£50–500K non-dilutive venture builder).
- The Challenge is a pedagogical exercise, not a spin-off mechanism.
- → Methodology transferable; structure is not (institution-bound, MBA/design students).

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

RCA/Imperial: 'flipping' Lean Startup

ORIGINAL SOURCE EXCERPT · p. 8

er 2009
2009
er 2009
009
2
19
9
19
009
, 2009

Blank, S. (2023), Lean Meets Wicked Problems, Steve Blank blog / Imperial College Business School class account.

COMMENTS

- Standard Lean Startup starts with a customer problem; deep tech starts with a technology capability and often NO existing market.
- WAL inverts the logic: technology-first speculative design → then validate with lead/extreme users.
- Endorsed by Steve Blank as a legitimate evolution for deep tech.
- → The key asset NCCR Muoniverse should adopt: the flipped methodology.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

CRL: the 9-point commercial scale

ORIGINAL SOURCE EXCERPT · p. 31

CRL 4	Minimum Viable Product Completed and Pilots initiated
CRL 5	Minimum Marketable Product and Operational Processes Validated
CRL 6	Minimum Marketable Product Deployed in the Market with Operating Revenue Targets
CRL 7	Matured Product Design and Marketing Strategy Validated
CRL 8	Matured Product Deployed in the Market with Targets Achieved
CRL 9	A Trusted Solution and a successful business model established

Market Readiness Levels (MRL) Overview and Rationale

NITI Aayog (2023), A New Lens for Innovation in New India: TCRM Matrix, p. 31.

COMMENTS

- Runs parallel to NASA TRL; with MRL forms three 'orthogonal' maturity axes.
- Key concept: the binding constraint — whichever axis lags blocks progress. Rule: CRL should always $\geq$ TRL.
- CRL 1–3 covers opportunity recognition → market sizing → competitiveness — conceptually relevant to TRL 1–3.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

CRL: the documentation problem

ORIGINAL SOURCE EXCERPT · p. 8

margin.

Taken together, TRL, MRL, and CRL create what we describe elsewhere in this series as three orthogonal axes of progress. They also highlight the nonlinear nature of Deep Tech progress. A company can be

Select "Accept all" to agree to our use of cookies and similar technologies to enhance your browsing experience, security, analytics and customisation. Select "Manage cookies" to make

COMMENTS

- Despite being one of the most-cited 'commercial readiness' concepts, CRL has NO peer-reviewed primary paper.
- Exists as proprietary consultancy IP; cited by Deep Tech Leaders, ITONICS, NITI Aayog.
- The '22_CRL_Abbas.pdf' in the evidence folder is a DIFFERENT EIT Urban Mobility variant — not Abbas & Nomvar.
- → Must be cited as a practitioner framework, not an academically validated instrument.

Deep Tech Leaders (2025), TRL, MRL & CRL Explained, Part 1, p. 8.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

CRL: the commercial dimension TRL misses

ORIGINAL SOURCE EXCERPT · p. 32

	MRL 2	Potential Business Models Formulat
	MRL 3	Early Stakeholders Identified and th
	MRL 4	Feedback from Pilot of Minimum V
	MRL 5	Market Sizing for Minimum Marketa Operating Revenue Targets Genera
	MRL 6	Problem Solution Fit Validated with Competitor Analysis Completed
	MRL 7	Feedback from Minimum Marketabl Research for Matured Product Com
	MRL 8	Product Market Fit Validated with F

COMMENTS

- No per-level evidence checklists, no rubric, no independent assessment protocol, no certification.
- Descriptive (where you are) not prescriptive (what to do next) — unlike IRL/VIRAL.
- Operationally thin at low levels: CRL 1 looks identical for a muon venture and a mobile app.
- → Best used as a diagnostic framing tool (a detector can be TRL 9 but CRL 1).

NITI Aayog (2023), TCRM Matrix, p. 32 (identifying the limits of TRL and the need for commercial readiness).

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse: the native Swiss framework

ORIGINAL SOURCE EXCERPT · p. 1



Innosuisse (official), Start-up Coaching overview.

COMMENTS

- ~CHF 300M/yr, ~20 instruments — the most institutionally relevant framework: any NCCR venture will interact with it.
- Coaching: Initial (CHF 10K/12 mo) → Core (CHF 50K/36 mo, milestone-gated) → Scale-up (CHF 75K/24 mo).
- Voucher system: startups choose their own accredited coaches.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse: bespoke, voucher-based coaching

ORIGINAL SOURCE EXCERPT · p. 2

Scale-up Coaching

Scale-up Coaching grants you a voucher worth up to CHF 75 000 for coaching sessions with Innosuisse accredited coaches over a period of up to 24 months.

Innosuisse (official), Start-up Coaching overview, voucher-based programme description.

COMMENTS

- Coaching methodology is NOT explicitly based on I-Corps or a formal Lean Startup curriculum.
- Mentor-driven, bespoke per startup — fundamentally different from I-Corps' standardized 7-week immersion.
- Entrepreneurship Training (venturelab, since 2004) IS curriculum-based — alumni include Climeworks, GetYourGuide.
- → A direct methodological contrast for the benchmarking: mentor-driven vs methodology-driven.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse: KOF-verified causal effect

ORIGINAL SOURCE EXCERPT · p. 21



Hulfeld, F., Spescha, A. & Wörter, M. (KOF ETH Zurich, 2024), Funding R&D Cooperation between Firms and Universities.

COMMENTS

- Only independently verified causal estimate: +20.7% sales, +17.6% employment over 5 years (matched control of 537 firms).
- Effects accelerate: employment +7.8% (yr 0) → +17.2% (yr 2) → +27.9% (yr 4).
- Strongest for small firms (5–50 employees) with ETH Domain research partners.
- The only framework in the benchmark with causal validation.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse support is stage-specific

ORIGINAL SOURCE EXCERPT · p. 2

Support offer Start-up Coaching

Initial Coaching

Initial Coaching grants you a voucher worth up to CHF 10 000 for coaching sessions with Innosuisse accredited coaches over a period of up to 6 months.

Innosuisse (official), Start-up Coaching overview — stage-specific support and durations.

COMMENTS

- No instrument designed for TRL 1–3 commercialization; BRIDGE PoC (12 months, CHF 130K) is too short for muon validation cycles.
- Coaching vouchers can't buy lab time, detector development, or validation experiments.
- Vanishingly few accredited coaches have scaled particle-physics ventures.
- Bottom-up model may not mobilize mission-oriented physics commercialization.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

EIC Accelerator: TRL 6–8

ORIGINAL SOURCE EXCERPT · p. 1

Develop and scaleup innovations

Develop and scaleup innovations with the potential to create or disrupt markets (TRL 6-8). Patient capital principle.

Startups and SMEs

Single start-ups and individuals intending launch a SME and si mid-caps (can apply equity only).

COMMENTS

- Flagship Accelerator requires TRL 5–8 — years away for emerging muon ventures.
- Blended finance (grant up to €2.5M + equity €0.5–10M); success rate 3–7% — among the most competitive globally.
- Relevant early-TRL pathway: Pathfinder (TRL 1–4, €3–4M grants) + Transition (TRL 3–6), both needing international consortia.

European Innovation Council (official), EIC Accelerator programme page.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

EIC: Switzerland fully eligible since Jan 2025

ORIGINAL SOURCE EXCERPT · p. 2

Documents

0 DECEMBER 2014

European Commission (official), International cooperation with Switzerland in research and innovation.

COMMENTS

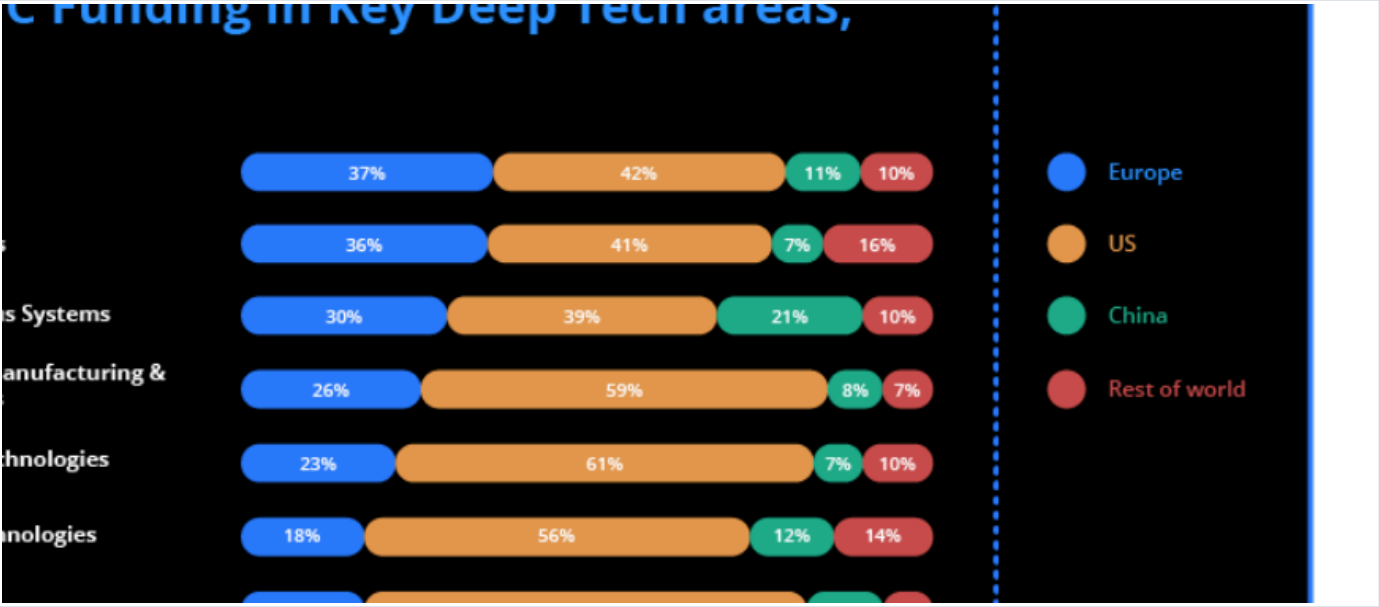
- Swiss firms fully eligible for all instruments incl. equity; can coordinate consortia.
- Feb 2026: 4 Swiss winners (Neutrality, Verity, Pregnolia, DePoly); June 2026: 4 of 38 — incl. BTRY (€2.2M) and Nanoflex (€12.5M).
- Swiss EIC winners are typically Venture Kick / Venture Leaders alumni — the national pipeline prepares them.
- → Sequencing: Innosuisse early → EIC Pathfinder/Transition → Accelerator → STEP Scale Up.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Candidate: CERN Venture Connect

ORIGINAL SOURCE EXCERPT · p. 3



Stryber (2025), Deep Tech commercialization in Europe — source identifying CERN Venture Connect as a relevant institutional actor.

COMMENTS

- CERN faces the IDENTICAL challenge as NCCR Muoniverse: fundamental particle physics → commercial applications.
- CVC model: 0% equity, express agreements, curated technology portfolio, 60 partner orgs across 18 countries.
- 8 startups joined in 2025, collectively raising >€5M.
- → The most directly comparable institutional model — must be benchmarked in Part 2.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

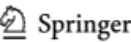
Candidate: NSF I-Corps

ORIGINAL SOURCE EXCERPT · p. 4

mentation and management and, where applicable, additional points of contact associated with regional or institution-specific interventions (e.g., individual universities participating in EXIST-Culture of Entrepreneurship awards). A total of 34 individuals were interviewed between 2023 and 2024 for this project, including program administrators, representatives from other interventions within relevant ecosystems, and (where applicable) researchers and

¹It is important to distinguish between the Lean Startup framework, introduced by Eric Ries (2008, 2011), which emphasizes iterative product development and validated learning, and the Lean LaunchPad methodology, developed by Steve Blank in 2011 as an experiential training model for scientists and engineers, later trademarked in 2014. We thank one of our reviewers for helping us differentiate this excellent point.

²Errol Arkilic, personal communication (2022). The relationship between the scientific method and Lean LaunchPad has also been recognized by scholars who frame the approach as part of broader conceptualizations such as “scientific entrepreneurship” (Camuffo et al., 2020).



COMMENTS

- The most internationally validated early-TRL commercialization program — adopted in 8 countries (EXIST Germany, Brazil, +6).
- Core methodology: lean startup + intensive customer discovery, structured 7-week immersion.
- Direct comparator to Innosuisse's bespoke coaching: methodology-driven vs mentor-driven.
- → STRONGLY RECOMMENDED for the November 1 benchmarking milestone.

Hayter, C. S. et al. (2025), Journal of Technology Transfer, p. 4.

The numbers

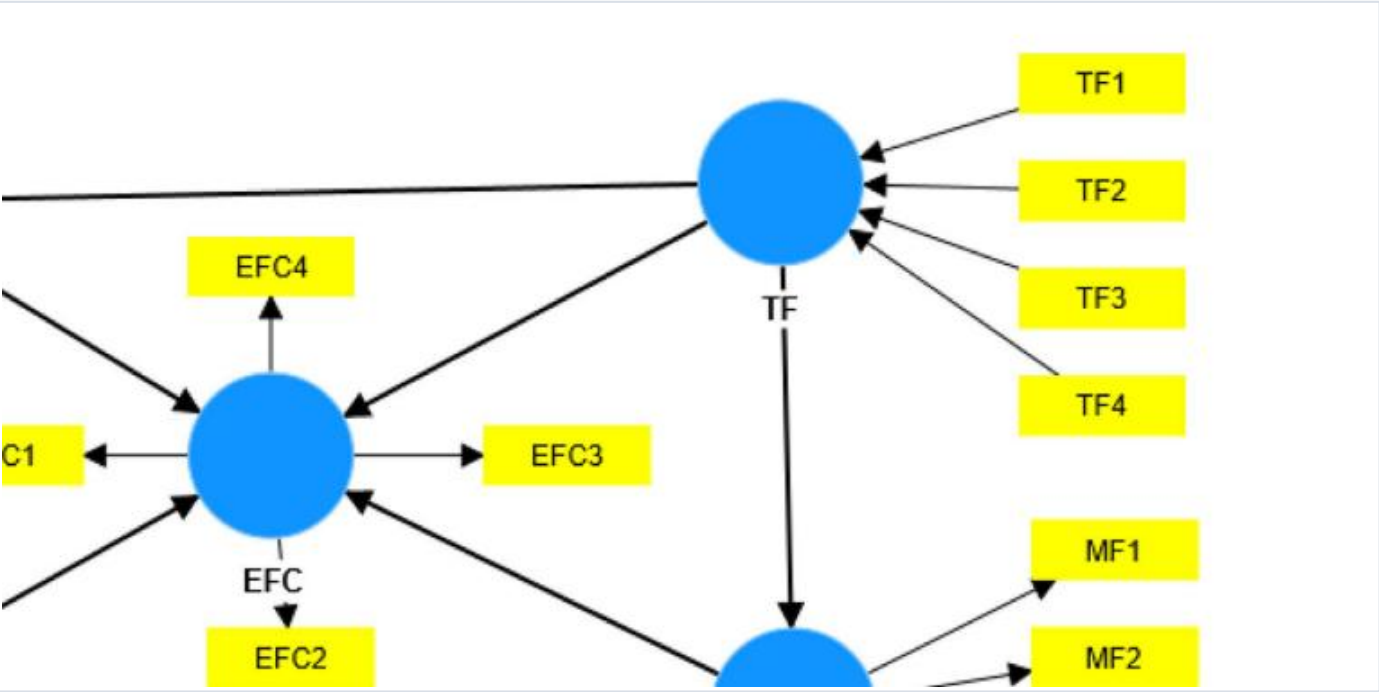
Statistical evidence behind the findings



SOURCE-LED FINDING

Market features dominate ($\beta = 0.514$)

ORIGINAL SOURCE EXCERPT · p. 10



Capatina, A., Bleoju, G. & Kalisz, D. (2024), Journal of Innovation & Knowledge 9, 100515, structural model.

COMMENTS

- Market features $\beta = 0.514$ — strongest predictor by far.
- Technology $\beta = 0.310$, environmental $\beta = 0.306$.
- Firm internal features: NOT significant.
- → Market identification is the single highest-leverage activity.

SOURCE-LED FINDING

Talent and knowledge drive deep tech

ORIGINAL SOURCE EXCERPT · p. 40

startups in the region	2.18 ***	2.00 2.39	1.90 ***	1.75 2.23	1.5 **
Generic EE index			1.02 *	1.00–1.04	0.9 *

Your privacy, your choice

We use essential cookies to make sure the site can function properly. We also use optional cookies to enhance your browsing experience, analyze site usage, and assist in our marketing efforts. You can manage your cookie preferences at any time.

COMMENTS

- Talent pools: IRR 1.66 — strongest predictor.
- Knowledge infrastructure: IRR 1.48.
- Formal institutions (policy, IP law): NOT significant.
- Top 10 regions = 40% of all deep tech startups.

Micol, F., Leendertse, J. & van Rijnsoever, F. (2026), Journal of Technology Transfer, Table 4.

DATA COMPENDIUM

SOURCE-LED FINDING

Incubation: type and age matter

ORIGINAL SOURCE EXCERPT · p. 32

<i>z</i>	<i>sig</i>	Q/QB	Prob≥Q	<i>r</i> ² (%)	<i>τ</i> ²
	*	355.08	0.0000	85.20	0.0263
		5.65	0.059		
	ns	195.56	0.0000	89.35	0.0434
	**	19.62	0.0204	58.96	0.0108
	***	42.84	0.0033	54.38	0.0074
		0.02	0.878		
	*	153.53	0.0000	78.74	0.0272
	ns	200.19	0.0000	90.18	0.0266
	**	337.62	0.0000	85.72	0.0268
		1.86	0.173		
	ns	292.35	0.0000	91.67	0.0397
	***	40.05	0.0211	43.63	0.0051
		1.34	0.247		
	ns	296.88	0.0000	90.07	0.0363
	***	39.51	0.0058	53.01	0.0067
		1.64	0.200		
	***	335.92	0.0000	86.20	0.0279
	**	0.49	0.4843	0.00	0.0000
		1.34	0.247		
	ns	296.88	0.0000	90.07	0.0363
	***	39.51	0.0058	53.01	0.0067

COMMENTS

- University incubators *r* = 0.118 vs overall 0.067; older (>20 yrs) *r* = 0.183; generic business incubators −0.011 (NS).
- Networking service *r* = 0.071; mentoring *r* = −0.012 (NS).
- Technology/R&D ventures benefit more (0.082–0.130).

Seitz, N. et al. (2026), Review of Managerial Science, moderator table.

SOURCE-LED FINDING

Advisors: business +0.23, technology -0.31

ORIGINAL SOURCE EXCERPT · p. 19

.1 Alternative endogeneity correction

our main analyses, we used entropy balancing to correct for intrinsic differences between
ial and novice entrepreneurs. As an alternative scheme, we used incorporation at the
e of the first survey as the treatment, reasoning that founders who legally incorporated
y be more serious and would expend more effort in growing them. We instrument this
ection with the same covariates used in the primary analysis. This treatment (Appendix
oles 4, 5) leads to two differences in our findings. First, while still negative, the main

like the coefficients of advice, those of experience reflect standardized units. The magnitude of effects
ussed here is for their unstandardized values.



COMMENTS

- Each business advisor: +0.23 AR units (p = 0.04).
- Each technology advisor: -0.31 AR units (p = 0.05).
- 4 business advisors ≈ +1 unit ≈ 15 yrs experience; 3 tech advisors ≈ -1 unit.
- 80%+ of ventures used at least one advisor — ubiquitous policy tool.

Denoo, L., Van Boxstael, A. & Belz, A. (2024), Journal of Technology Transfer, p. 1585.

SOURCE-LED FINDING

TTOs become playmakers through relationships

ORIGINAL SOURCE EXCERPT · p. 16

6.1 Implications for practice

TTOs' important contributions should be adequately resourced and overseen in line with environmental, social and institutional goals and strategies. Government investment in joint knowledge ventures (e.g., tax credits and grant fund matching with TNCs) will support innovation network building, successful commercialisation projects, and enhance ANZ's reputation in the global life sciences arena. TTO-led government-TNC collaboration may increase TNC engagement in ANZ. Investment in a physical presence at international industry conferences is important. Research funding allocation should support both basic and



COMMENTS

- The primary case study describes TTOs as 'playmakers' whose influence grows through relational ties and access to the wider innovation-service ecosystem.
- The key mechanism visible in the source is relationship-building, not a narrow technology-transfer transaction.
- → The interview phase should test whether Swiss TTOs operate this way in practice.

Chen, Z., Little, V. & Thuan, N. (2025), Journal of Technology Transfer, p. 1075.

DATA COMPENDIUM

SOURCE-LED FINDING

Switzerland: #1 per capita, 3× the runner-up

ORIGINAL SOURCE EXCERPT · p. 3



Deep Tech Nation Switzerland / Dealroom.co (2025), Swiss Deep Tech Report 2025, via Swiss Startup Association reproduction.

COMMENTS

- Spinout value per capita index: 100 vs 33 (runner-up).
- ETH #3, EPFL #4, UZH #8 in Europe.
- 60–63% of Swiss VC flows to deep tech — #1 globally (China 56%, US 54%).
- 3 of 6 global >\$1B spinout exits in 2025 were Swiss (Araris, Nextthink, u-blox).

SOURCE-LED FINDING

Innosuisse Innovation Projects: 57% launch rate

ORIGINAL SOURCE EXCERPT · p. 2

Comparative impact results

Selected monitoring results, which are collected in a standardised way, allow comparison made across the support offers.

- **Satisfaction:** Funding beneficiaries are satisfied or very satisfied with the support provided by Innosuisse and their implementation: 89% of Innovation Boosters, around 90% of researchers and 80% of implementation partners among implementation partners, and 68% of start-ups in Core Coaching. The number of funding recipients who are not satisfied or only slightly satisfied with the funding offer is consistently low, from 1% to 2%.
- **Incremental and radical innovations:** Innosuisse funds both radical and incremental innovations. The respective shares differ depending on the support offer. In innovation projects with implementation partners, 25% of innovations are classified as radical.

COMMENTS

- 57% market launch within 3 years post-funding.
- 2.5 highly qualified jobs per project (~800–900 FTE per cohort).
- Leverage: 1 CHF → 4 CHF value added.
- ~50% continue with the research partner after 3 years.

Innosuisse (2025), Impact monitoring 2021–2023.

SOURCE-LED FINDING

BRIDGE PoC: 70% founding rate

ORIGINAL SOURCE EXCERPT · p. 3

Innovation cheque

The innovation cheque offers SMEs and start-ups low-threshold funding. The aim is to carry out feasibility and market studies as well as technological clarifications.

- **Cooperation with research:** Thanks to the innovation cheque, more than half of the participating companies can be introduced to knowledge and technology transfer with research. 70% of SMEs continue to work with their research partner after the project is completed.
- **Results achieved:** The collaboration with research partners is successfully tested in almost all cases. 63% of companies have successfully tested and further developed their innovation idea and fully achieved this goal. Around a quarter of innovation cheques subsequently carry out an innovation project.

Innosuisse (2025), Impact monitoring 2021–2023, BRIDGE Proof of Concept.

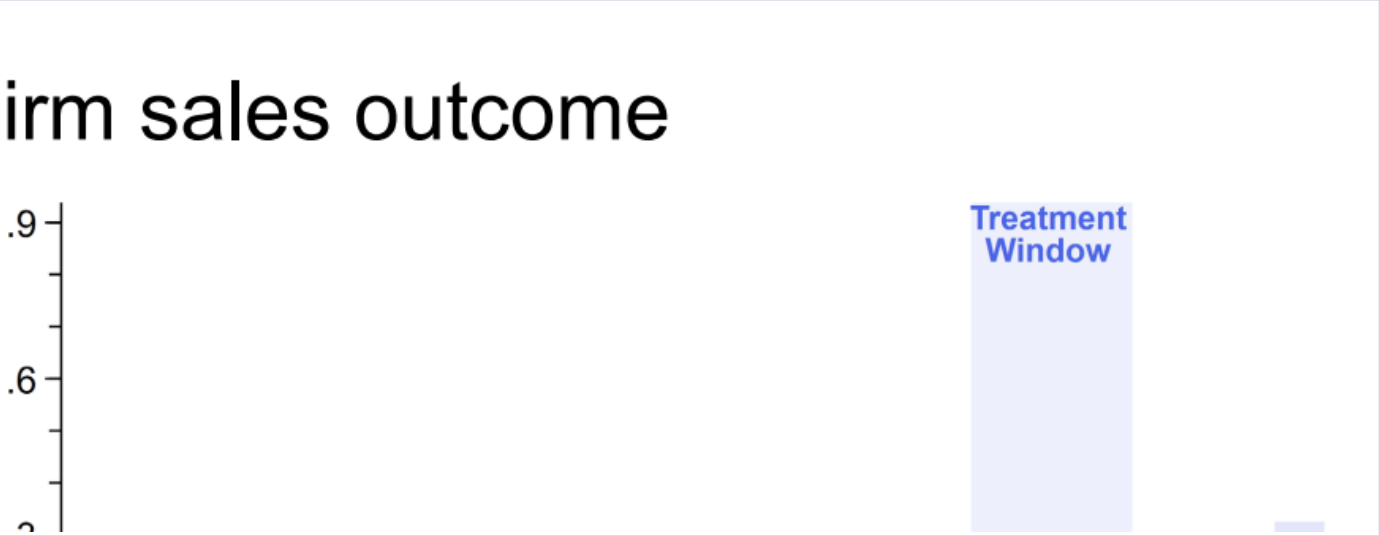
COMMENTS

- 70% of funded researchers go on to found a startup — highest conversion of any Innosuisse instrument.
- CHF 130K over 12 months; 80 startups created from 121 PoC projects.
- → Proves the concept works when the timeline fits — muon timelines do not.

SOURCE-LED FINDING

KOF causal effect: +20.7% sales, +17.6% employment

ORIGINAL SOURCE EXCERPT · p. 21



COMMENTS

- +20.7% sales and +17.6% employment over 5 years vs matched controls.
- Employment effect grows: +7.8% (yr 0) → +17.2% (yr 2) → +27.9% (yr 4).
- Statistically significant only for small firms (5–50 employees) with ETH Domain partners.

Hulfeld, F., Spescha, A. & Wörter, M. (KOF ETH Zurich, 2024), causal difference-in-differences result.

SOURCE-LED FINDING

The TRL 1–3 gap, visualized

ORIGINAL SOURCE EXCERPT · p. 30

	TRL 3	Proof-of-Concept
	TRL 4	Technology Component / sub-system validation in laboratory environment
	TRL 5	Technology Component / sub-system validation in relevant environment (industrially relevant environment in case of l enabling technologies)
	TRL 6	Technology sub-system or prototype demonstration in a relevant environment
	TRL 7	Technology System Prototype demonstration in an operational environment
	TRL 8	Actual Technology System completed and qualified through testing and demonstration

NITI Aayog (2023), TCRM Matrix, p. 30 (TRL scale).

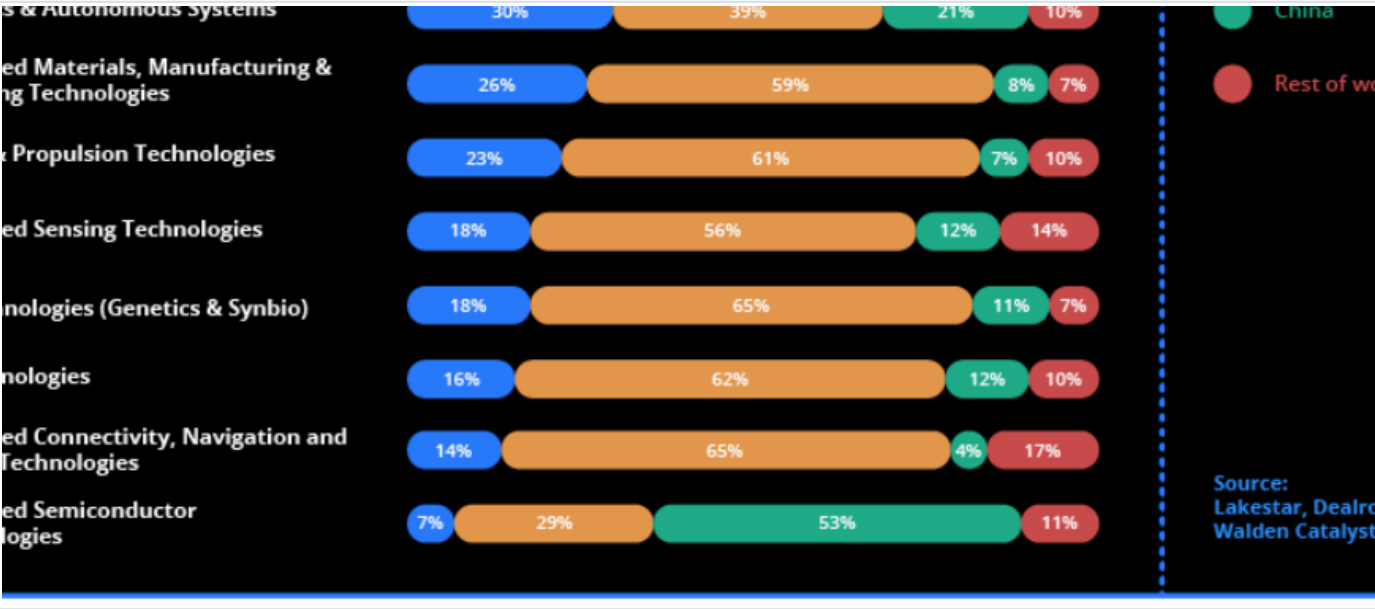
COMMENTS

- RCA/Imperial: TRL 4–6. CRL: 1–9 (conceptual only). Innosuisse: 4–7. EIC: 1–8 pipeline.
- Red zone = TRL 1–3 — where NCCR Muoniverse operates.
- Only Innosuisse has independent causal validation (KOF DiD); only EIC publishes an assessment methodology.

SOURCE-LED FINDING

EU vs US: a 3.7× capital gap

ORIGINAL SOURCE EXCERPT · p. 3



Stryber (2025), Deep Tech commercialization in Europe.

COMMENTS

- Deep tech VC: €58B (EU) vs €215B (US) — 3.7× gap.
- 62% of EU investors reject failed founders vs 23% US (2.7× more risk-averse).
- PhD-to-startup: 15% vs 45%; corporate engagement: 12% vs 40%.
- US holds 49% of global deep tech VC.

DATA COMPENDIUM

SOURCE-LED FINDING

Deep tech: +35% time, +48% capital

ORIGINAL SOURCE EXCERPT · p. 54

other challenge is the structural gap in funding mechanisms. While VC remains the dominating source, many deep tech startups struggle to secure early-stage financing, particularly capital-intensive projects like space tech and advanced computing. Non-dilutive funding sources such as government grants, can play a key role in supporting early innovation, but accessing funds is often complex and highly competitive, as some stakeholders highlighted in the interview. The grant application process can involve administrative burdens, rigid milestone-l

UNDP (2025), Global Deep Tech Ecosystems, p. 54.

COMMENTS

- Time to milestones: 135 vs 100 (+35%).
- Capital required: 148 vs 100 (+48%).
- Development timeline: 1–3 yrs conventional vs 5–15+ yrs deep tech.
- 97% of deep tech ventures contribute to ≥1 SDG.

MILESTONE 1

Source-led evidence → interview questions

The practical conclusion: early-TRL physics commercialization is not missing ideas — it is missing a support architecture designed for the gap.

Next: Milestone 2 interviews · September 1, 2026

All source excerpts are embedded from the primary PDFs/pages listed beneath each image.